Number of ν_e appearance events

$$N = P(\nu_\mu \to \nu_e; E_0, L)\, \sigma_{CC}(E_0)\, \Phi\, \mathrm{POT}\, N_T \;\simeq\; 60.2\, P \left[\frac{\sigma_{CC}}{10^{-38}\,\mathrm{cm}^2}\right] \left[\frac{\Phi}{10^{-14}\,\mathrm{cm}^{-2}\,\mathrm{POT}^{-1}}\right] \left[\frac{\mathrm{POT}}{10^{21}}\right] \left[\frac{M}{\mathrm{kt}}\right]$$

The second form follows from writing $N_T = M\, N_A$ (one nucleon per atomic mass unit, $N_A = 6.022 \times 10^{23}\,\mathrm{g}^{-1}$) and measuring each factor in the units in brackets: the constant is $10^{-38} \cdot 10^{-14} \cdot 10^{21} \cdot (10^9\,\mathrm{g/kt} \cdot 6.022 \times 10^{23}\,\mathrm{g}^{-1}) = 60.2$, and all units cancel. N is the number of ν_e ($\bar{\nu}_e$) charged-current interactions in the fiducial volume, i.e., the number of events seen by a perfect detector (efficiency = 1). The last section (*Why the experiments see a different number*) explains why the experiments see a different number.

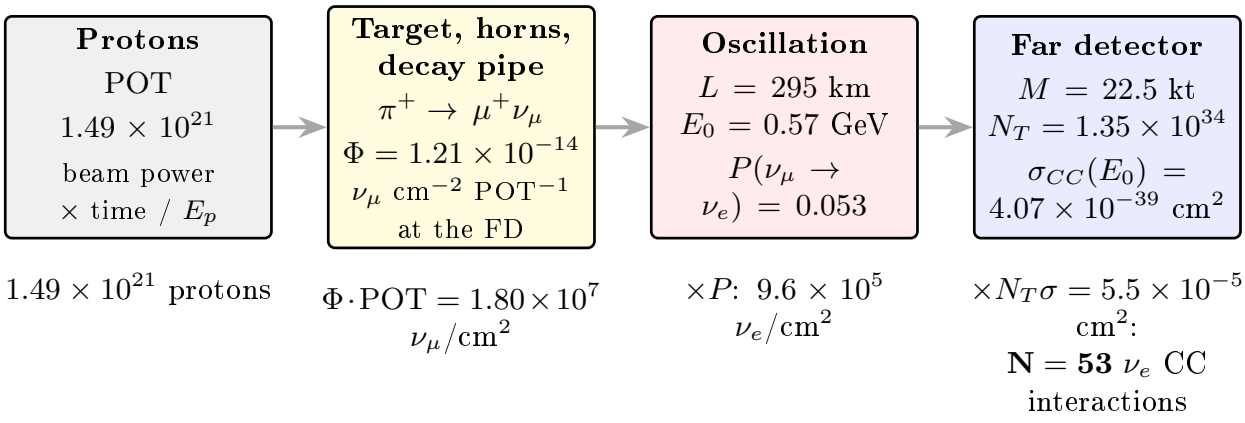


Figure 1: The chain of the calculation for the T2K neutrino beam (NuFIT 6.0, normal ordering). Each block is one factor of the formula; below each block, the running result. Protons on target produce, through the target, the focusing horns and the decay pipe, a ν_μ beam: at the far detector it gives Φ neutrinos per cm^2 per POT. A fraction P oscillates into ν_e over the baseline L; the far detector offers a target area $N_T\, \sigma_{CC}$.

- $\Phi \times \mathrm{POT}$ is the **beam side**: the number of ν_μ crossing each cm^2 of the far detector (how the beam is produced: Fig. 2). $\Phi = \int (d\Phi/dE)\, dE = \sum_k (d\Phi/dE)_k\, \Delta E_k$ is the ν_μ ($\bar{\nu}_\mu$) flux per proton on target *integrated over the whole energy spectrum* (not taken at E_0), summing the bins of the public flux histograms (T2K: Phys. Rev. D 87, 012001 (2013); NOvA: FERMILAB-DATA-2017-01; DUNE: arXiv:2103.04797), see Fig. 3. POT is the exposure, POT = beam power × time/proton energy.
- $\sigma_{CC} \times N_T$ is the **detector side**: the target area seen by a neutrino. $N_T = M N_A$ is the number of nucleons in the fiducial mass M; σ_{CC} is the inclusive charged-current cross section per nucleon (GLoBES tables, Comput. Phys. Commun. 167, 195 (2005); GENIE on argon for DUNE, arXiv:2103.04797), see Fig. 4.
- P is the oscillation probability in matter (NuFast, arXiv:2405.02400), here at the NuFIT 6.0 best fit (JHEP 12 (2024) 216). It is the only factor that depends on the oscillation parameters: varying δ_{CP}, the mass ordering or θ_{23} changes N only through P, all other factors fixed.
- E_0 replaces the energy spectrum by a single energy, the mean energy of the beam peak.
- **Assumptions**: pure ν_μ ($\bar{\nu}_\mu$) beam, one energy, every interaction detected, no backgrounds, constant matter density. The formula gives no energy spectrum.

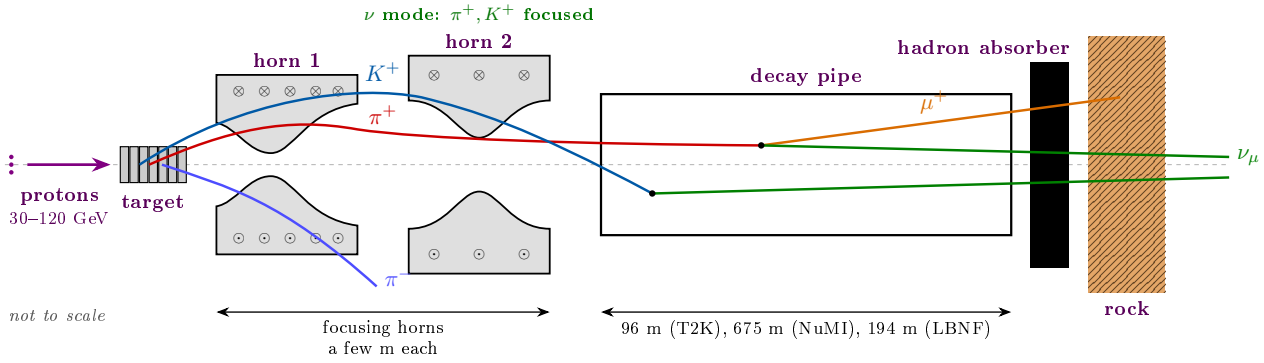


Figure 2: Schematic of an accelerator neutrino beam in neutrino mode (not to scale). Protons hit a graphite target and produce mesons. Two magnetic horns, with a toroidal field between the inner and outer conductors, focus the positive mesons (π^+, K^+) towards the beam axis and defocus the negative ones (π^-). The mesons decay in the decay pipe, mainly $\pi^+ \to \mu^+\nu_\mu$; the absorber and the rock stop the remaining hadrons and the muons, and only the neutrinos continue to the detectors. Reversing the horn current focuses π^-, K^- and gives the $\bar{\nu}_\mu$ beam. Decay pipe lengths: 96 m at T2K, 675 m at NuMI (NOvA), 194 m at LBNF (DUNE).

Inputs (NuFIT 6.0 normal ordering for P).

	beam	L [km]	M [kt]	POT	E_0 [GeV]	Φ [10^{-14}/cm^2/POT]	σ_{CC} [10^{-38} cm^2]
T2K	ν	295	22.5	1.494×10^{21}	0.57	1.207	0.407
	$\bar{\nu}$			1.635×10^{21}	0.55	0.976	0.116
Hyper-K	ν	295	187	6.75×10^{21}	0.57	1.207	0.407
	$\bar{\nu}$			2.025×10^{22}	0.55	0.976	0.116
NOvA	ν	810	14	2.661×10^{21}	1.80	1.806	1.43
	$\bar{\nu}$			1.25×10^{21}	1.79	1.467	0.631
DUNE	ν	1300	40	7.15×10^{21}	2.31	1.548	1.78
	$\bar{\nu}$			7.15×10^{21}	2.27	1.385	0.729

Worked out ($N_{P=1} = 60.2\,\sigma\,\Phi\,\mathrm{POT}\,M$ in the units above, then $\times P$).

	beam	$N_{P=1}$	$N = N_{P=1} \times P$
T2K	ν	$60.2 \times 0.407 \times 1.207 \times 1.494 \times 22.5 = 994$	$\times 0.0533 = \mathbf{53}$
	$\bar{\nu}$	$60.2 \times 0.116 \times 0.976 \times 1.635 \times 22.5 = 251$	$\times 0.0316 = \mathbf{7.9}$
Hyper-K	ν	$60.2 \times 0.407 \times 1.207 \times 6.75 \times 187 = 37\,329$	$\times 0.0533 = \mathbf{1990}$
	$\bar{\nu}$	$60.2 \times 0.116 \times 0.976 \times 20.25 \times 187 = 25\,809$	$\times 0.0316 = \mathbf{816}$
NOvA	ν	$60.2 \times 1.43 \times 1.806 \times 2.661 \times 14 = 5792$	$\times 0.0547 = \mathbf{317}$
	$\bar{\nu}$	$60.2 \times 0.631 \times 1.467 \times 1.25 \times 14 = 975$	$\times 0.0249 = \mathbf{24}$
DUNE	ν	$60.2 \times 1.78 \times 1.548 \times 7.15 \times 40 = 47\,441$	$\times 0.0738 = \mathbf{3501}$
	$\bar{\nu}$	$60.2 \times 0.729 \times 1.385 \times 7.15 \times 40 = 17\,384$	$\times 0.0178 = \mathbf{309}$

Sensitivity to CP violation. Each experiment measures two numbers, N_ν and $N_{\bar{\nu}}$. The agreement between observed numbers N_i^{obs} and the prediction N_i^{pred} of a hypothesis is measured with

$$\chi^2 = \sum_{i=\nu,\bar{\nu}} \frac{\left(N_i^{\mathrm{obs}} - N_i^{\mathrm{pred}}\right)^2}{\sigma_i^2}, \qquad \sigma_i^2 = N_i^{\mathrm{obs}},$$

since the number of events is Poisson distributed and its statistical variance equals the number of events (no normalization uncertainty is included). For the expected sensitivity: (i) the observed numbers are the prediction for the true value, $N_i^{\mathrm{obs}} = N_i(\delta_{CP})$, without fluctuations, so the best fit is the true value itself, with $\chi^2 = 0$; (ii) CP conservation means $\delta_{CP} = 0$ or π: both must be rejected, so we keep the

one that fits best; (iii) $\Delta\chi^2$, the difference from the best fit ($\chi^2 = 0$), is then

$$\boxed{\Delta\chi^2(\delta_{CP}) = \min_{\delta' \in \{0,\pi\}} \sum_{i=\nu,\bar{\nu}} \frac{[N_i(\delta_{CP}) - N_i(\delta')]^2}{N_i(\delta_{CP})}, \qquad \text{significance} = \sqrt{\Delta\chi^2}\,\sigma}$$

with N_i from the formula above (only P changes with δ_{CP}). For one parameter, $\Delta\chi^2 = 9$ (25) corresponds to 3σ (5σ) (PDG, Review of Particle Physics, Statistics chapter). The CP-violation sensitivity is defined, as in the DUNE (arXiv:2002.03005) and Hyper-K (arXiv:1805.04163) design studies, as $\sqrt{\Delta\chi^2}$ with respect to $\delta_{CP} = 0, \pi$ computed on Asimov data. Our version is a simplified counting version: Gaussian statistics, two samples, no backgrounds, fixed oscillation parameters. With the numbers of the table (true normal ordering, statistical errors only), the maximum significance over δ_{CP} is 2.5σ (T2K), 4.4σ (NOvA), 17σ (Hyper-K) and 18σ (DUNE). Derivation and the case with a normalization uncertainty: `docs/number_of_events.pdf`.

Why the experiments see a different number

The collaborations publish the number of appearance signal events expected in their analyses, computed with full simulations of beam, cross sections and detector. Evaluating our formula with the same oscillation parameters, the ratio

$$r = \frac{N_{\text{published}}}{N}$$

would be 1 if the formula were exact. Table 1 shows that it is not: $r = 0.4$–1.1. Writing the published number as an integral over the spectrum, r splits into two factors,

$$r = \underbrace{\frac{N_{\text{published}}}{\text{POT}\, N_T \int \Phi(E)\,\sigma(E)\,P(E)\,dE}}_{\varepsilon_{\text{det}}} \times \underbrace{\frac{\int \Phi(E)\,\sigma(E)\,P(E)\,dE}{\Phi\,\sigma(E_0)\,P(E_0)}}_{c_{\text{mono}}},$$

which correspond to the two simplifications of the formula:

- **Detection** (ε_{det}): not every ν_e interaction is selected (detector efficiency and analysis cuts), so $\varepsilon_{\text{det}} < 1$. In practice ε_{det} also absorbs the differences between our cross sections and the generators used by the collaborations.
- **One energy** (c_{mono}): the beam is not monochromatic. In the ν beams E_0 is close to the oscillation maximum, while many neutrinos have energies where P is smaller: the formula overestimates the rate, $c_{\text{mono}} \approx 0.8$. In the $\bar{\nu}$ beams the antineutrino cross section grows quickly with energy, so the high-energy part of the flux weighs more: $c_{\text{mono}} \approx 1$ at T2K/Hyper-K and 1.4 at DUNE.

So $r < 1$ in the ν beams, where both effects reduce the rate, and $r \approx 1$ or even > 1 in the $\bar{\nu}$ beams of Hyper-K and DUNE, where the one-energy underestimate compensates the efficiency. r is therefore not an efficiency: it collects everything the formula leaves out. (It is not due to the beam contamination: the references are right-sign signal only.)

Consequences. Everything that depends only on P (bi-probability plots, shape of the δ_{CP} dependence, degeneracies) is unchanged. The numbers of events of the formula are those of a perfect detector: about twice the published ones for NOvA, within ∼40 % for the other experiments. The significances are optimistic: multiplying each sample by its r, the maximum sensitivity to CP violation (statistics only, true normal ordering) becomes 2.3σ (T2K), 2.9σ (NOvA), 16σ (Hyper-K) and 16σ (DUNE). With a 5 % normalization uncertainty the Hyper-K and DUNE values (8.6–9.4σ) hardly change, because they are limited by the systematic uncertainty and not by the number of events.

Table 1: Comparison with the published predictions, at the oscillation parameters of each reference. T2K: Table XI total single-ring e-like events at $\delta_{CP} = -\pi/2$ minus the δ_{CP}-independent background obtained by fitting the four δ_{CP} values of the table.

	beam	N (formula)	$N_{\text{published}}$	r	c_{mono}	$\varepsilon_{\text{det}} = r/c_{\text{mono}}$	reference
T2K	ν	61.6	54.8	0.89	0.81	1.10	arXiv:2101.03779, Table XI
	$\bar{\nu}$	6.7	4.8	0.71	1.06	0.68	arXiv:2101.03779, Table XI
Hyper-K	ν	2064	1643	0.80	0.85	0.94	arXiv:1805.04163, Tables XXXVI–XXXVII
	$\bar{\nu}$	1156	1183	1.02	1.00	1.02	arXiv:1805.04163, Tables XXXVI–XXXVII
NOvA	ν	290.5	125.3	0.43	0.79	0.54	arXiv:2509.04361, Table 2
	$\bar{\nu}$	37.2	18.9	0.51	0.83	0.61	arXiv:2509.04361, Table 2
DUNE	ν	3486	2180	0.63	0.78	0.79	arXiv:2103.04797, GLoBES configuration
	$\bar{\nu}$	409	438	1.07	1.39	0.77	arXiv:2103.04797, GLoBES configuration

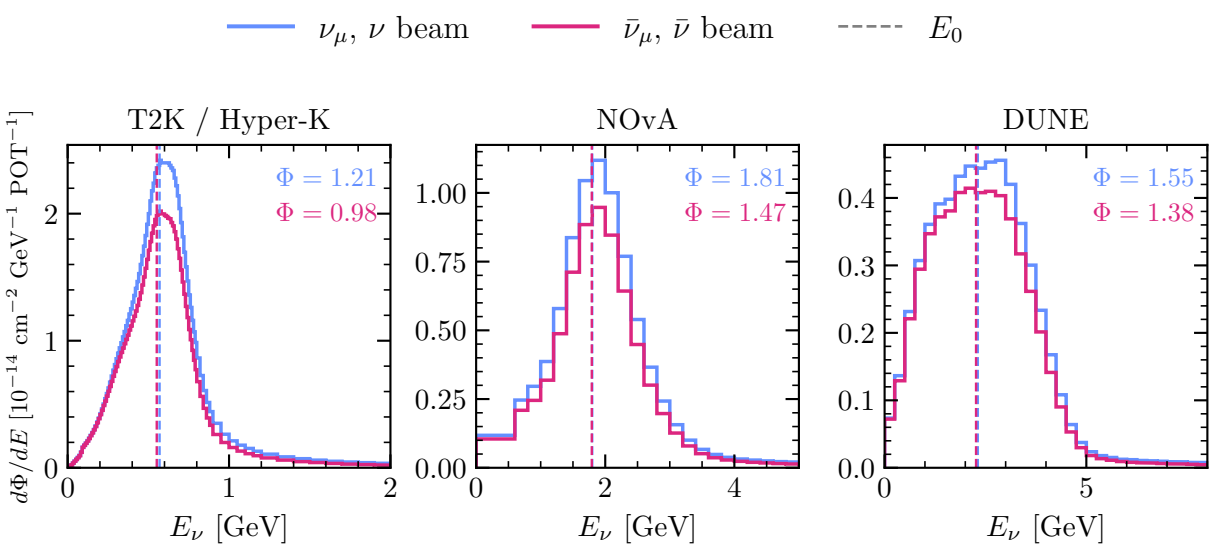


Figure 3: Unoscillated ν_μ flux in the neutrino beam and $\bar{\nu}_\mu$ flux in the antineutrino beam at the far detectors, per proton on target, from the public flux releases (T2K: 2020 release at Super-Kamiokande, also used for Hyper-K; NOvA: 2017 near-detector flux $\times\,(1\,\text{km}/810\,\text{km})^2$; DUNE: TDR far-detector flux). Dashed lines: the beam energy E_0 (flux-weighted mean energy of the main peak). The numbers give $\Phi = \sum_k (d\Phi/dE)_k\,\Delta E_k$ in units of 10^{-14} cm^{-2} POT^{-1}, as used in the table.

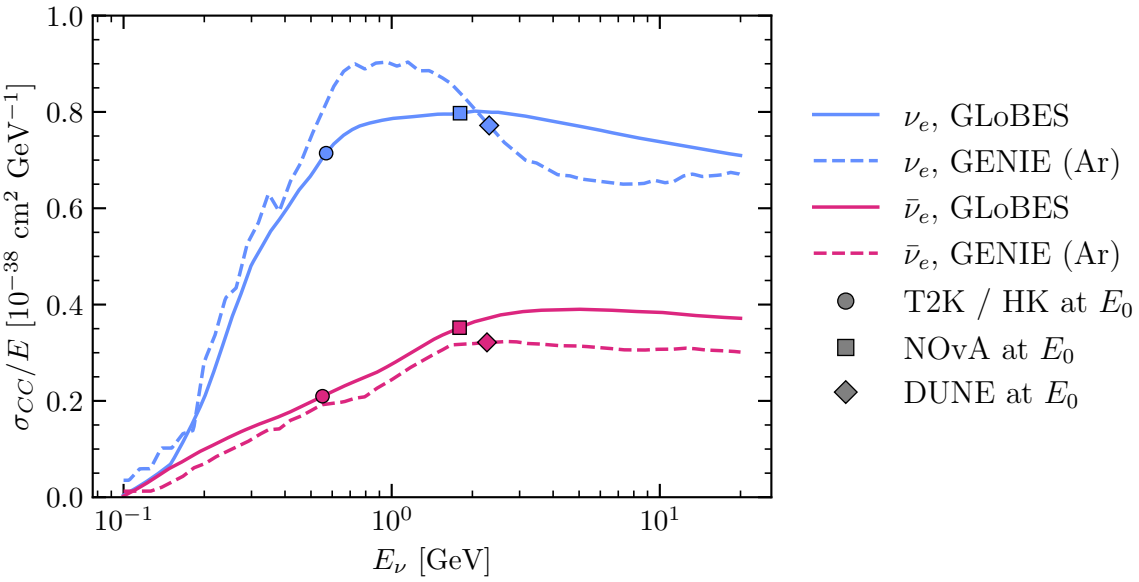


Figure 4: Inclusive charged-current cross section per nucleon, divided by energy, for ν_e and $\bar{\nu}_e$: GLoBES standard table (low energy: Super-Kamiokande cross sections for water, M. D. Messier, PhD thesis (1999); high energy: Paschos and Yu, Phys. Rev. D 65, 033002 (2002)), used for T2K, Hyper-K and NOvA; GENIE on argon of the DUNE TDR configuration (arXiv:2103.04797), used for DUNE. Markers: $\sigma_{CC}(E_0)/E_0$ entering the table.